Volume V · Chapter 24 — Perception for Clinical/Surgical Scenes · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-V-Chapter-24-context.md` for the AI interaction log.

Prerequisites: Ch 23; Ch 6–7 (vision-capable models).

Learning outcomes — the student can: apply computer vision to surgical/clinical scenes; segment/track instruments & anatomy; estimate depth/pose; integrate perception into a robot loop (in simulation).

Labs (hands-on) — 14 h:

#	Lab	h
24a	Segment instruments/targets in a simulated surgical scene	4
24b	Track an instrument across frames	4
24c	Depth/pose estimation on sim data	3
24d	Close a simple perception→action loop in sim	3

Datasets/tools: OpenCV, segmentation models; the simulator. **Assessment:** perception pipeline (**60%**); tracking report (**20%**); quiz (**20%**). **Key decisions:** model choice vs. latency; sim-to-real gap; **safety of perception errors**. **References:** plan §7; §13 → *Robotics & embodied AI*; *Medical LLMs* (vision). **Hours:** Theory **8** + Lab **14** = **22**.